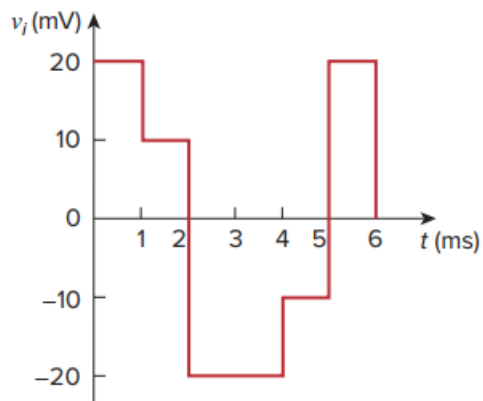


Problem

An op amp integrator with $R = 4 \text{ M}\Omega$ and $C = 1 \text{ }\mu\text{F}$ has the input waveform shown in Fig. 6.88. Plot the output waveform.

Fig. 6.88:



Step-by-step solution

Step 1 of 2

$$RC = 4 \times 10^6 \times 1 \times 10^{-6} = 4$$

$$V_o = \frac{-1}{RC} \int V_i dt = \frac{-1}{4} \int V_i dt$$

$$\text{For } 0 < t < 1, \quad V_i = 20, \quad V_o = \frac{-1}{4} \int_0^t 20 dt = -5t \text{ mV}$$

$$\begin{aligned} \text{For } 1 < t < 2, \quad V_i = 10, \quad V_o &= -\frac{1}{4} \int_1^t 10 dt + V(1) = -2.5(t-1) - 5 \\ &= -2.5t - 2.5 \text{ mV} \end{aligned}$$

$$\begin{aligned} \text{For } 2 < t < 4, \quad V_i = -20, \quad V_o &= \frac{1}{4} \int_2^t 20 dt + V(2) = 5(t-2) - 7.5 \\ &= 5t - 17.5 \text{ mV} \end{aligned}$$

Step 2 of 2

For $4 < t < 5$, $V_i = -10$, $V_0 = \frac{1}{4} \int_4^t 10 dt + V(4) = 2.5(t-4) + 2.5$

$= 2.5t - 7.5 \text{ mV}$

For $5 < t < 6$, $V_i = 20$, $V_0 = \frac{-1}{4} \int_5^t 20 dt + V(5) = -5(t-5) + 5$

$= -5t + 30 \text{ mV}$

Thus $V_0(t)$ is As shown below.

